2.2 CHI-SQUARE DISTRIBUTION AND ITS APPLICATIONS



Karl Pearson

Karl Pearson (1857-1936) was a English Mathematician and Biostatistician. He founded the world's first university statistics department at University College, London in 1911. He was the first to examine whether the observed data support a given specification, in a paper published in 1900. He called it 'Chi-square goodness of fit' test which motivated research in statistical inference and led to the development of statistics as separate discipline.

Karl Pearson chi-square test the dawn of Statistical Inference - *C R Rao.*

Karl Pearson's famous chi square paper appeared in the spring of 1900, an auspicious beginning to a wonderful century for the field of statistics - *B. Efron*

2.2.1 Chi-Square distribution

The square of standard normal variable is known as a chi-square variable with 1 degree of freedom (d.f.). Thus

If $X \sim N(\mu, \sigma^2)$, then it is known that $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$. Further Z^2 is said to follow χ^2 – distribution with 1 degree of freedom (χ^2 – pronounced as chi-square)

Note: i) If $X_i \sim N(\mu, \sigma^2)$, $i = 1, 2, \ldots, n$ are n *iid* random variables, then

$$\sum_{i=1}^{n} Z_i^2 = \sum_{i=1}^{n} \left(X_i - \mu\right)/\sigma^2 \text{ follows } \chi^2 \text{ with } n \text{ } d.f \text{ (additive property of } \chi^2)$$

ii) If μ is replaced by $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ then $\frac{\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2}{\sigma^2}$ follows χ^2_{n-1}

2.2.2 Properties of χ^2 distribution

- It is a continuous distribution.
- The distribution has only one parameter *i.e. n d.f.*
- The shape of the distribution depends upon the *d.f*, *n*.
- The mean of the chi-square distribution is n and variance $2n$
- If U and V are independent random variables having χ^2 distributions with degree of freedom n_1 and n_2 respectively, then their sum $U + V$ has the same χ^2 distribution with *d.f* $n_1 + n_2$.

2.2.3 Applications of chi-square distribution

- To test the variance of the normal population, using the statistic in note (ii) (sec. 2.2.1)
- To test the independence of attributes. (sec. 2.2.5)
- To test the goodness of fit of a distribution. (sec. 2.2.6)
- The sampling distributions of the test statistics used in the last two applications are approximately chi-square distributions.

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